

Newborough Warren

Why the Water Under the Dunes Matters for Wildlife

A condensed summary of a 21-year groundwater study

Hollingham, M. (2026). Hydrogeological Dynamics, Behavioural Clustering and Management Intervention Analysis at Newborough Warren Coastal Sand Dune Aquifer, Wales. Draft, 2026.

Project: github.com/newbroman/Newborough_Hydrology | Contact: martin.hollingham+nrg@gmail.com | ORCID: 0000-0003-0253-9301

The place

Newborough Warren sits at the southern tip of Anglesey — 1,300 hectares of sand dunes, wetlands and conifer plantation, designated as a Special Area of Conservation. The heart of the site is its dune slacks: shallow, seasonally flooded hollows between the dunes that are among the rarest habitats in Britain. They support nationally scarce plants including Fen Orchid (*Liparis loeselii*), creeping willow and a rich community of wetland invertebrates that exist nowhere else in Wales at this density. All of them depend on the water table sitting within 37 cm of the surface through the summer. That margin — about the length of a ruler — is the difference between a thriving wet slack and an irreversibly degraded dry one. A 700-hectare Corsican pine plantation, planted between 1948 and 1965, intercepts roughly a quarter of incoming rainfall before it can reach the water table. This study monitored 88 dipwells monthly for 21 years (2005–2026), asking a single ecological question: is the water table staying close enough to the surface to keep the dune slack communities viable?

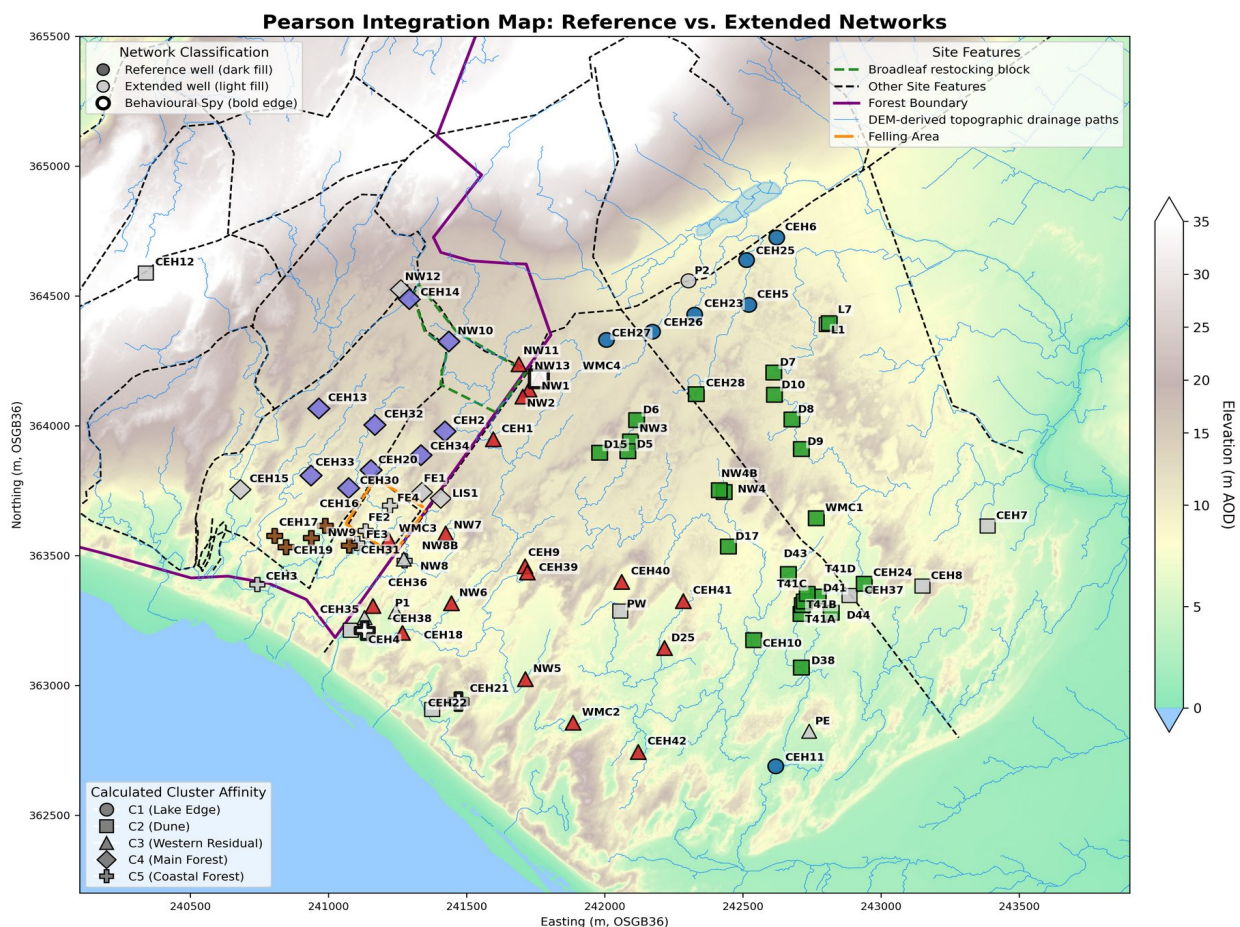


Figure 1. Spatial mapping of the five zones: C1 Lake Edge (blue), C2 Dune (green), C3 Western Residual (red), C4 Main Forest (purple), C5 Coastal Forest (brown).

Five groundwater zones — five ecological situations

A water balance model fitted to each well groups the site into five zones with distinct ecological outlooks. C1 (Lake Edge) is the most immediately threatened zone — shallow, fast-draining, and already recording individual summers where the water table reaches the wet-slack viability limit. C2 (Dune) supports the bulk of the open dune slack habitat; moderate rainfall sensitivity gives it some resilience but long-term decline is measurable. C3 (Western Residual) acts as a slow deep buffer, responding less strongly to individual dry summers but not immune to the long-term trend. C4 (Main Forest) sits on thin sand over irregular bedrock: pine interception and a low-capacity substrate suppress winter recharge; the clearfell is intended to reverse this. C5 (Coastal Forest) is declining fastest of all, driven by coastal retreat as well as pine interception. The key ecological point is that the thickness of sand beneath each area — not just tree cover — determines how severely evaporation draws down the water table in summer.

An independent test against botanical survey data confirms the model captures ecologically relevant behaviour: the equilibrium wetness state computed from the fitted coefficients predicts plant community moisture indicator values (Ellenberg-F) across eighteen monitored wells as accurately as the directly measured spring-level baseline ($r = 0.81$ vs $r = 0.83$, $n = 18$), but in only 30 months compared to MSL5's 60 months.

Zone	Character	Recharge sensitivity	Evaporation draw	Drainage rate	Canopy share
C1 Lake Edge	Shallow, fast-draining	4.58 (highest)	0.96 (lowest)	0.090 (fastest)	0.22
C2 Dune	Open mature dune	3.98	1.77	0.07	0.26
C3 W. Residual	Deep buffer	3.57	1.85	0.06	0.28
C4 Main Forest	Pine on thin bedrock substrate	2.52 (lowest)	2.55 (highest)	0.020 (slowest)	0.4
C5 Coastal Forest	Pine on deeper coastal sand	2.41	1.32	0.05	0.41

Table 1. The five groundwater zones. How much rain reaches the water table (recharge sensitivity), how hard evaporation pulls it down (evaporation draw), how quickly it drains away (drainage rate), and what fraction of the sky is shaded by canopy (canopy share). Higher recharge and lower evaporation draw is better for dune slack ecology.

What the data show

Summer is the critical season

The water table has a critical season and a critical depth. Summer is when plants and invertebrates experience maximum stress: if the water table drops below 61 cm in summer (the Curreli et al. 2013 wet-slack threshold), hygrophilous species cannot survive. If it drops below 98 cm, even dry-slack communities become marginal. The summer minimum has been declining across all five zones. The Lake Edge zone — the zone most directly linked to the orchid populations — is on course to cross the wet-slack threshold as a long-term mean around 2030–2032; individual dry summers already reach it. This is not a distant projection: the window for effective intervention is the next decade. UKCP18 projections indicate summer minima will deepen by a further 71–134 mm by the 2080s. The five-year mean spring baseline (MSL5), used to track management progress, is more stable — projected to deepen by 21–39 mm over the same period, three to five times less than the summer minimum. This divergence reflects the ecological asymmetry of the aquifer: winter recharge is partially buffered by drainage storage, but summer draw-down is not.

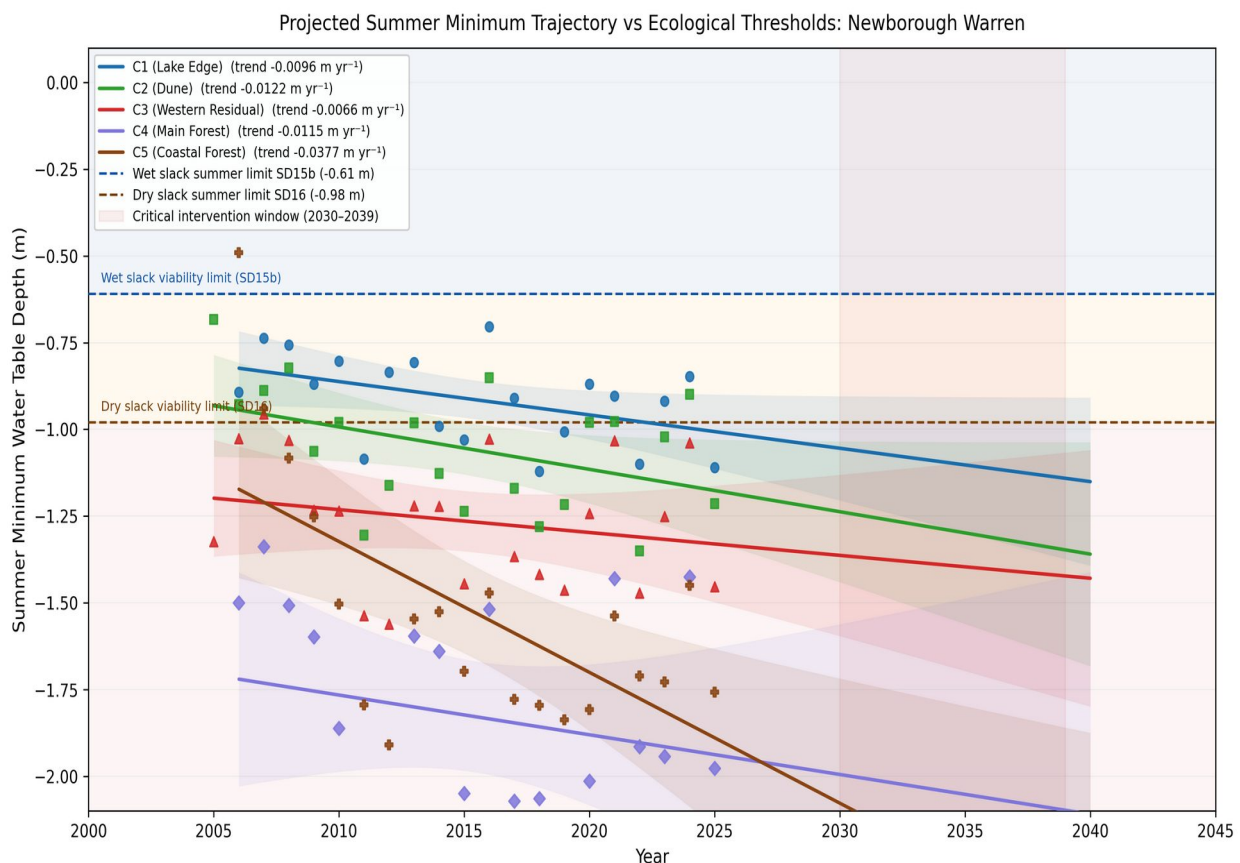


Figure 2. Projected summer minimum depth for all five zones. Wet-slack threshold -0.61 m (blue dashed); dry-slack threshold -0.98 m (brown dashed). Critical intervention window 2030–2039 shaded.

Dune scraping

Scraping removes several centimetres of sand from the slack floor, permanently lowering the ground surface and bringing it closer to the water table. The ecological effect is immediate and geometric: a slack scraped 20 cm deeper sits 20 cm closer to the water table for as long as the scrape geometry is maintained. At CEH36 (scraped April 2015), three independent analytical methods converge on a scraping benefit of 80–140 mm in

mean water-table depth; the ecologically relevant test — the shift in summer minimum depth against the unscrapped control well — was +195 mm ($p = 0.004$). That shift moved CEH36 from a zone where the wet-slack threshold was routinely exceeded to one where it is only reached in the driest years. The absolute water-table level at CEH36 has continued to decline since 2015 alongside the rest of the network; the improvement is maintained as a relative advantage over the unscrapped condition, not as a reversal of the long-term trend, and it is confined to the scraped ground itself — the raised water table is a property of the lowered surface and does not extend beyond the scrape edge. CEH18 and CEH21 were scraped in October 2023 at more seaward locations and have so far shown no detectable benefit in under two years of record; both sites are also subject to coastal-retreat boundary lowering that works against the intervention.

There is a physical side-effect of scraping, and the monitoring record now shows it directly. When a slack is scraped, the lowered ground surface acts as a local drain, pulling groundwater toward it from the surrounding area and drawing the water table down at neighbouring ground. At WMC3, a well 260 m from the CEH36 scrape, the water level stepped down relative to the unaffected forest by about 55 mm after the 2015 scraping — and by a near-identical amount again after the 2023 works elsewhere in the network (Figure 4). The consistency of that drop across two separate scraping events is what a genuine drainage effect would produce. The drainage response in this part of the aquifer is quick enough to see within the record: the water table here half-recovers from a disturbance in roughly 13 to 21 months, so a scrape leaves a mark on its neighbours within a year or two rather than over decades. This is a level effect on surrounding wells, distinct from the benefit at the scrape itself, and it is consistent with drainage toward the lowered surface rather than a confirmed measurement of how far the drawdown reaches. The scraping benefit at CEH36 is real and large; this is not an argument against scraping. It is an argument for spacing scraping events far enough apart that each draws from its own volume of aquifer, rather than compounding the drawdown on the slacks in between.

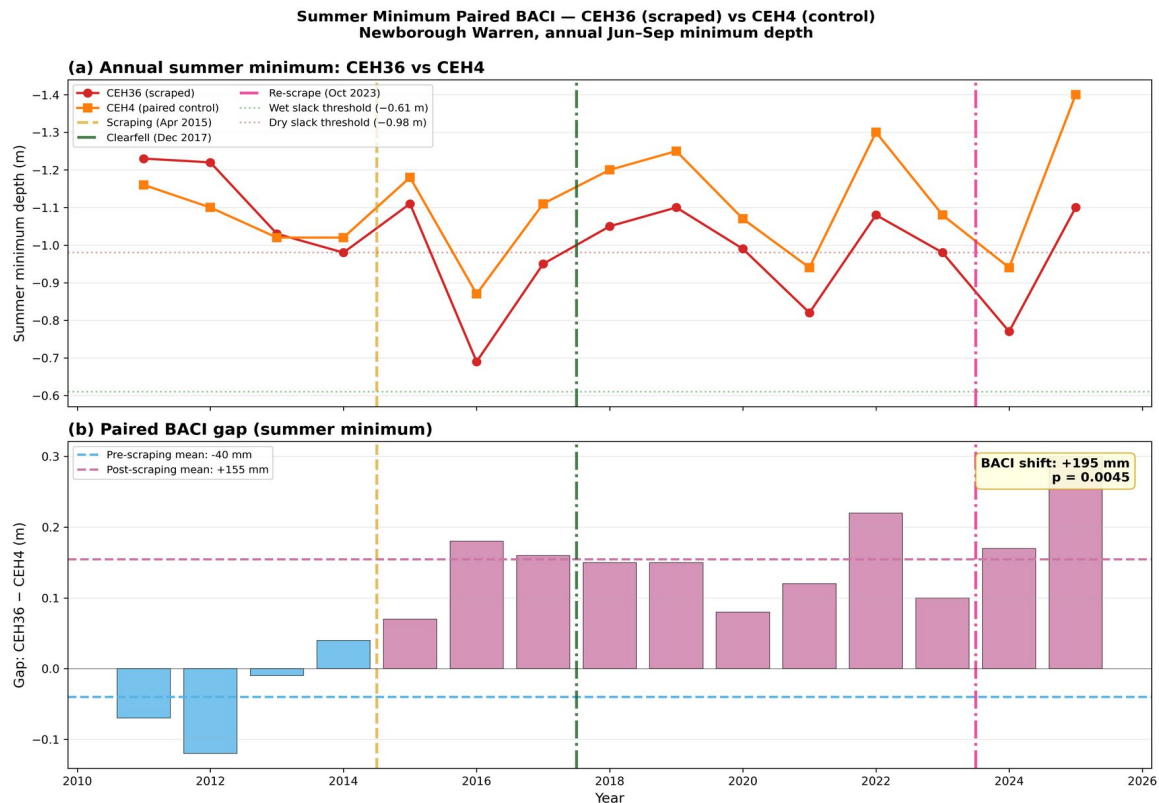


Figure 3. CEH36 (scraped, red) vs control CEH4 (orange). Paired BACI shift: +195 mm ($p = 0.004$). Top: annual summer minima. Bottom: gap between wells before and after scraping.

Tree felling

The December 2017 clearfell of 8.4 ha of Corsican pine produced a statistically significant rise in mean monthly water levels at the core impact well of +120 mm relative to unfelled forest controls ($p < 0.001$; Figure 4). The rise is visible as a sustained widening of the gap between the felled well and the unfelled forest after December 2017. Measured against the ecological yardstick that matters, however, the result is more nuanced. Summer minimum depths — the metric that determines whether dune slack communities can persist — showed no corresponding improvement across any post-felling year. The clearfell raised the annual average water level but produced no detectable benefit at the time of year that determines dune slack survival. The explanation lies in the dual role of the canopy: pine trees intercept rainfall in winter (reducing recharge) but also shade the soil in summer (reducing evaporation). Removing the canopy recovers some winter recharge but increases summer drying by a comparable amount, so the net effect on the summer minimum is near-zero. A further constraint is spatial: the hydraulic divide routes forest groundwater south-westward toward Caernarfon Bay, not eastward toward the Lake Edge and Dune zones where most of the slack ecology sits. Clearfell benefits are largely confined to within the former forest footprint.

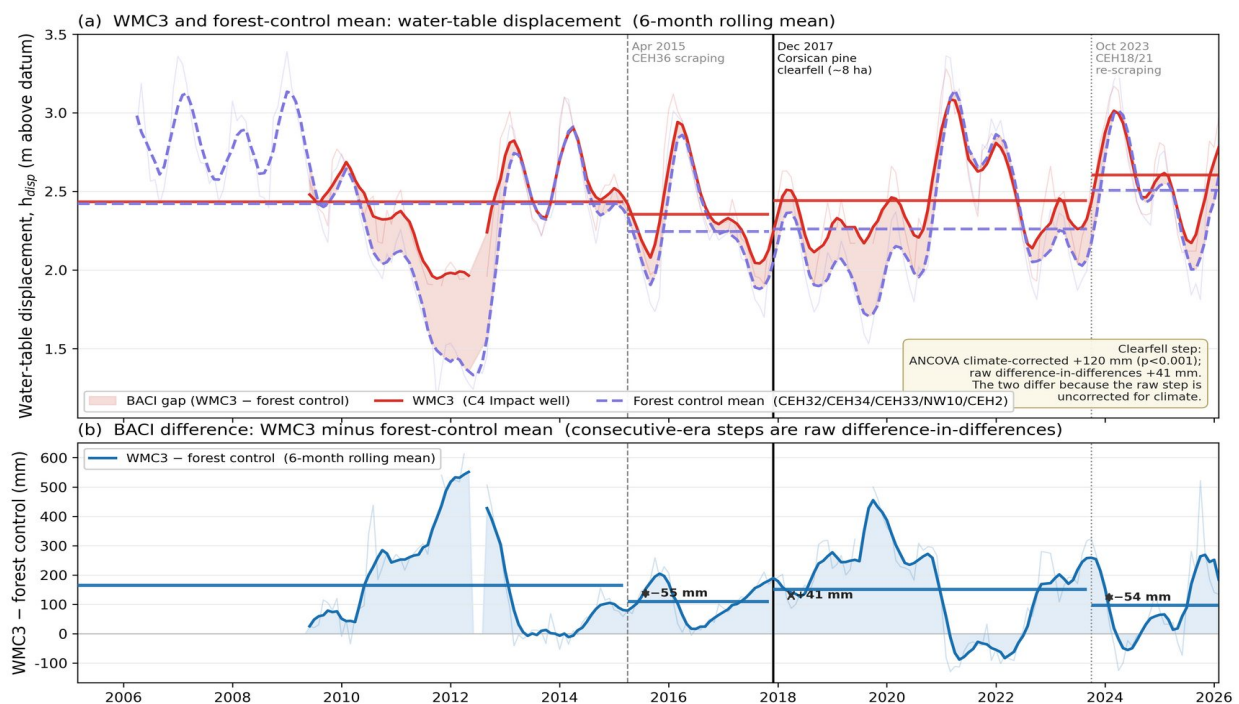


Figure 4. The felled well (WMC3, red) against the unfelled forest average (purple dashed), water levels relative to a fixed datum; shaded band is the gap between them. The gap widens after the December 2017 clearfell (+120 mm) and steps down at each scraping event (-55 mm in 2015, -54 mm in 2023) as the scrape draws the neighbouring water table down. Source: 10m_02_wmc3_baci_dual.png.

A site-wide recharge decline

A decline in recharge efficiency — the proportion of rainfall that reaches the water table — operates across all five zones regardless of land cover or management history. It is consistent with changing rainfall patterns and represents a climate-driven background trend against which all management interventions must be measured. Coastal retreat is a separate and spatially focused driver: as the shoreline erodes, the aquifer loses volume at its seaward edge and the water table adjusts downward across an area extending several hundred metres inland. Because groundwater moves slowly through sand, the declines visible today partly reflect erosion from several years ago. The coastal-retreat signal accounts for approximately 41% of the Coastal Forest zone — exceptional decline — a zone that cannot be restored by scraping or felling alone.

How water levels have changed

The next three maps show the same twenty years three ways, from the most direct measure to the most interpreted — separating what happened from why. Read together they are one argument in three steps, not three competing answers.

The most direct measure is what the water table actually did. Comparing the pre-clearfell five-year spring baseline (2013–2017) with the most recent comparable window (2019–2023) (Figure 5), the site-wide mean deepened by 97 mm: of 59 wells with data in both windows, 56 deepened by more than 25 mm and none became shallower by more than 25 mm. Ecologically, the average dune slack is starting each winter 97 mm worse than before the clearfell — a deficit winter recharge must overcome before any flooding benefit reaches slack vegetation. But this absolute change bundles together two things a manager needs to separate: the year-to-year weather, and the slow secular drivers underneath it.

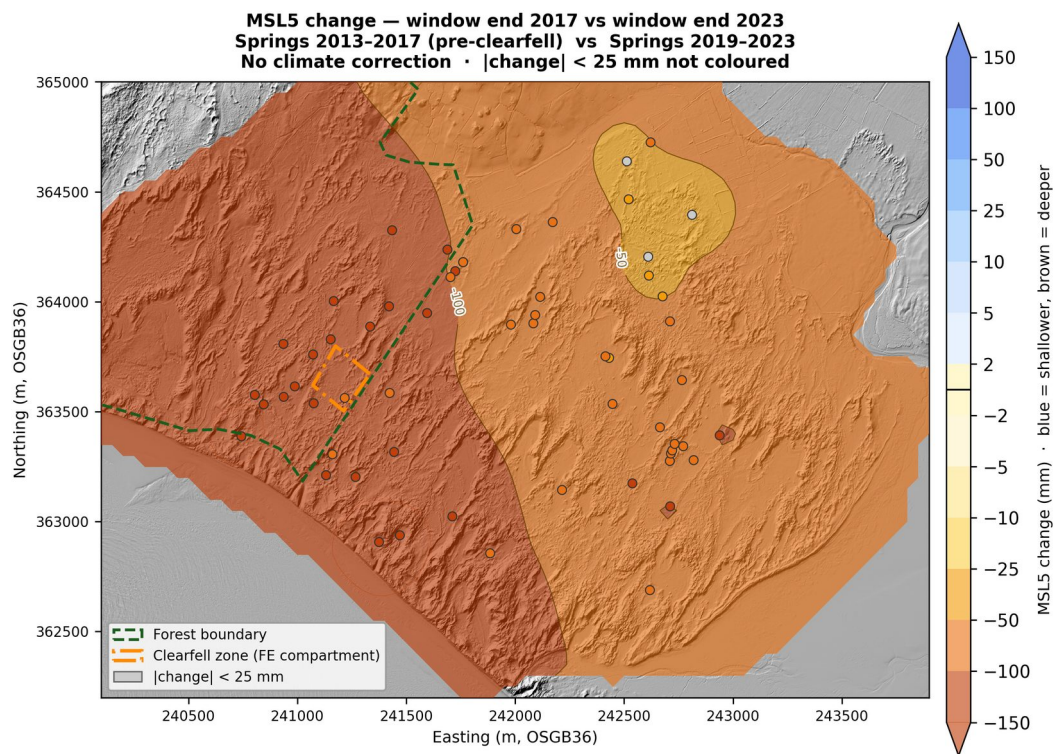


Figure 5. Observed MSL5 change: window end 2017 (springs 2013–2017, pre-clearfell) vs window end 2023 (springs 2019–2023). Raw difference in mm (positive = shallower); no climate correction. This is the absolute change — weather and long-term drivers combined. Source: 20_msl5_change_perwell.png.

Removing the common climate signal — the shared wet and dry years that move every well together — isolates the genuine long-term trend at each well, without re-referencing it to the site average (Figure 6). The picture that remains is spatial. The coastal fringe is genuinely drying: the Coastal Forest zone is losing roughly 12 mm per year once the weather is stripped out, and the western block is mildly drying. The interior — including the Main Forest — is close to holding its position. This corrects a tempting misreading of the raw record: the recent run of wet springs makes the forest look as if it is recovering, but that is amplification of the wet years, not a genuine long-term gain. The real, weather-independent losses are concentrated at the western and coastal margins.

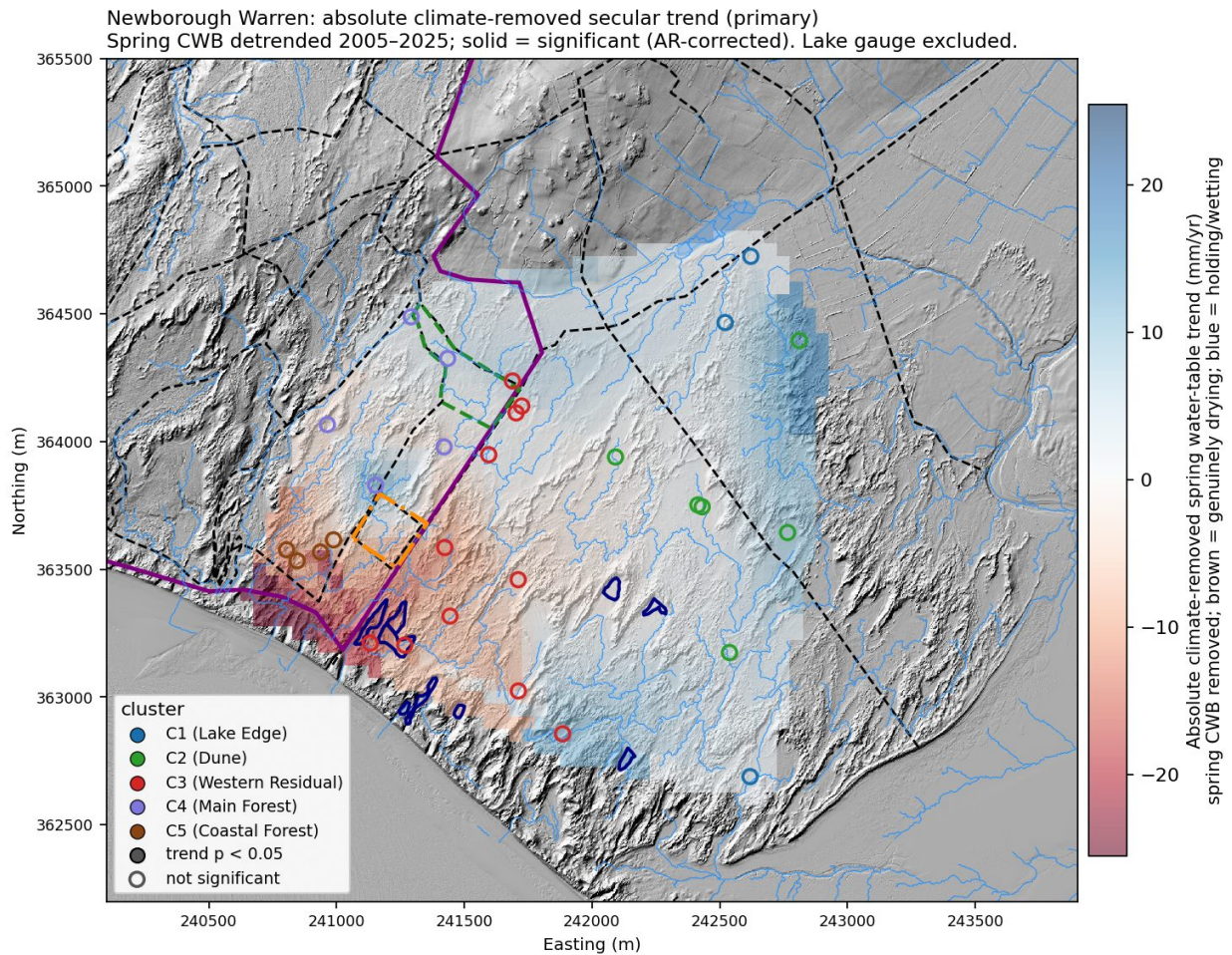


Figure 6. Absolute climate-removed water-table trend, 2005–2025. Each well's spring trend after the common weather signal (spring climatic water balance) is removed and NOT re-referenced to the network mean; brown = genuinely drying, blue = holding or wetting. The coastal and western fringe are the genuine driers; the interior holds once the weather is taken out. Source: 36_absolute_climate_trend_2005_2025.png.

What could drive that pattern? Figure 7 maps what the known drivers say should have changed between 2005 and 2025 — the modelled counterpart to the observed trend, built as the difference between the aquifer's driver state in 2005 and in 2025. The standing pine canopy, in place since the 1960s, cancels: it was drawing the water table down just as hard in 2005 as in 2025, so it adds nothing to the change over this period. What remains are the things that actually changed during the study: the 2017 clearfell (a local gain where the canopy was removed), the broadleaf restocking block (a local loss as its canopy re-established), the dune scrapes (local rises), and — dominating the western and southern margins — twenty years of coastal retreat and a small contribution from sea-level rise. The modelled coastal drying lines up with the observed climate-removed losses at the coast, tying the two maps together: the western fringe is losing ground because the shoreline is retreating, which outweighs the benefits of scraping or felling inland. Scraping at the coast only adds to the effects of coastal the drawdown in the unscraped areas.

Newborough Warren — MODELLED driver change, 2005 → 2025 **Clearfell gain · Broadleaf restock · Dune scrapes · 5-yr chronic coastal drawdown · SLR**

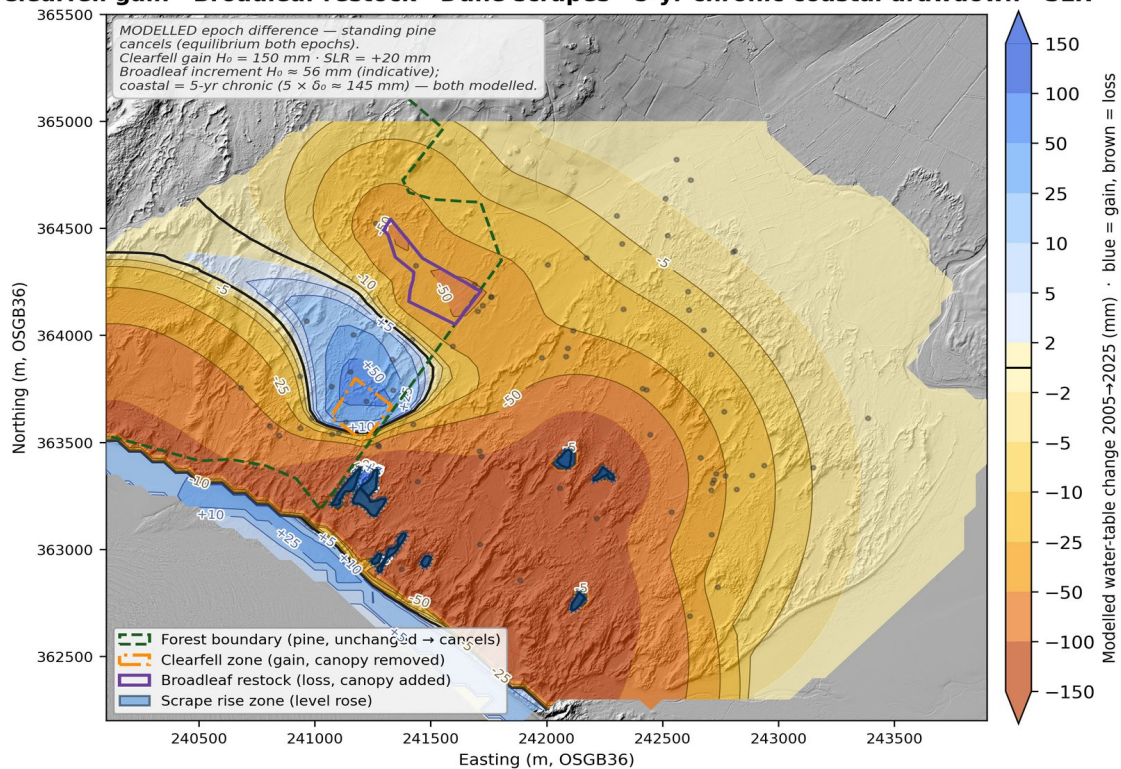


Figure 7. Modelled driver change, 2005 → 2025: what the known drivers say should have happened, as the difference between the aquifer's state in 2005 and in 2025. The standing pine canopy cancels — it was drawing the water table down equally in both years — so only the things that changed appear: the clearfell gain (blue), the broadleaf restock loss (brown, canopy re-established), the dune scrapes (local rises), and the coastal margin's five-year chronic drawdown with sea-level rise (brown, western and southern edges). The broadleaf and coastal fields are modelled and indicative, not fully resolved by the record. Blue = gain, brown = loss. Source: 20_driver_change_2005_2025.png.

The scale of observed change in context

The management interventions studied to date have produced measurable effects at the local scale: the scraping benefit at CEH36 is statistically robust and ecologically meaningful, having shifted that well's summer minimum by 195 mm — enough to move it from regular threshold-breaching to only occasional threshold-breaching. The clearfell produced a detectable improvement in mean monthly water levels within the former forest footprint. These are genuine, quantified achievements.

The spatial extent of each effect explains the asymmetry. The scrape benefit is confined to the scraped ground itself — the raised water table is a property of the lowered surface and does not extend beyond the scrape edge. Around it, the scrape instead draws the water table down, a drawdown that decays with distance and becomes small beyond roughly 225 m; the -55 mm fall observed at WMC3, 260 m from the scrape, sits on this modelled decay curve. The forest canopy drawdown decays over a comparable distance. Coastal retreat behaves differently: the shoreline drawdown decays linearly and continues to reach inland to roughly 900 m. A single 6 m retreat event comparable to the 2014 storm Brendan removes a shallow but site-wide head from the entire western dune field. Over five years, accumulated coastal retreat imposes a drawdown comparable in total magnitude to the standing forest canopy — but spread across the entire western aquifer rather than confined to a single zone (Figure 8).

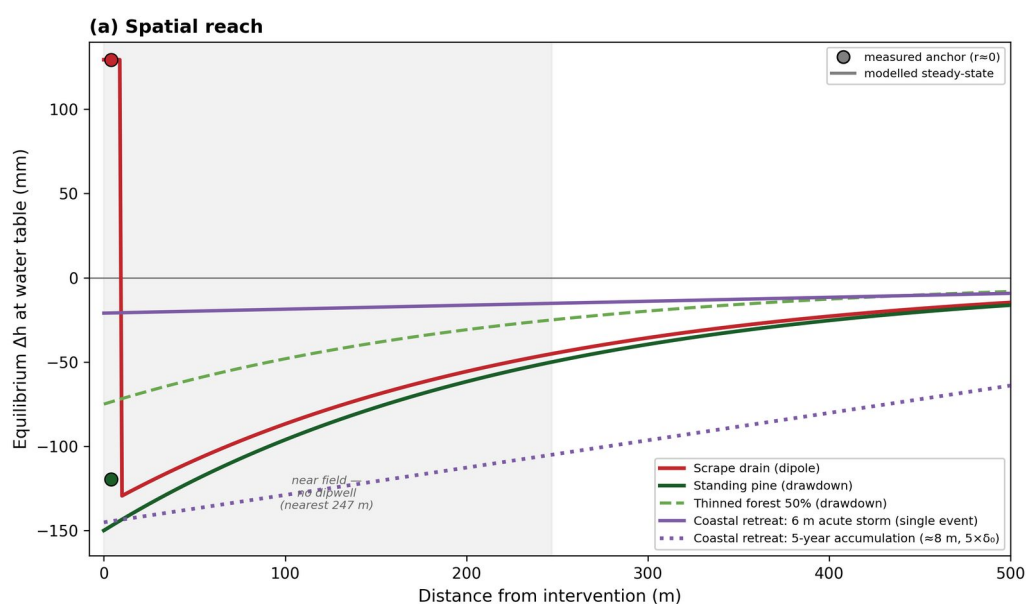


Figure 8. Spatial reach of management interventions and coastal retreat. Equilibrium head change as a function of distance from each source. Scrape benefit and forest drawdown decay exponentially (length scale ≈ 225 m); coastal-retreat drawdown decays linearly to ≈ 900 m. The two coastal curves differ in construction: the 6 m curve is a single storm-class event; the 5-year curve is five years' accumulation of the fitted retreat trend. All curves are modelled steady-state constructions anchored at two measured points (scrape BACI response; clearfell recovery). Source: 09f_management_effects.png.

The challenge is one of scale and rate. The site-wide spring baseline deepened by 97 mm between the 2017 and 2023 comparison windows — a change affecting 56 of 59 monitored wells simultaneously, driven by forces operating at the scale of the whole aquifer. Summer temperatures have trended upward at $+0.014$ °C yr^{-1} since 1931, with a step increase of $+0.94$ °C above baseline since 2013. UKCP18 projections indicate a further summer minimum deepening of 71–134 mm by the 2080s. Added to the 97 mm already lost between the 2017 and 2023 comparison windows, cumulative losses from the pre-clearfell baseline are projected to reach 170–230 mm — comfortably exceeding any management effect observed in this record, and operating simultaneously across the entire network rather than at a single location.

Interactive tools and further information

Three tools are freely available at newbroman.github.io/Newborough_Hydrology. The Scenario Viewer explores how UKCP18 climate projections and management options shift water levels across all five zones. The Flood Forecaster converts a current dipwell reading into a probability of winter flooding, updated through the season with live Met Office rainfall. The Seasonal Extremes Scatter plots every well against the Curreli et al. (2013) ecological viability thresholds. The full technical report, methods supplement and user manuals are available at the same address.